

A disproof of Conjecture 00000003848

The affine cell-core dictionary

1 The conjecture

Conjecture 00000003848 reads:

Definition: Two-sided cells of the affine 0-Hecke monoid are its two-sided equivalence classes. *Conjecture:* Two-sided cells of type $A_n^{(1)}$ correspond one-to-one with $(n+1)$ -core partitions, and the number of left cells in the corresponding cell equals the number of distinct parts of the partition obtained by removing all $(n+1)$ -hooks from that core.

We show it is false. For every affine rank $n \geq 1$ (the Lean statement also covers the formal case $n = 0$, which is harmless), the empty partition is an $(n+1)$ -core with 0 distinct parts, while every two-sided cell contains at least one left cell. For $n = 1$ we also give a non-degenerate refutation. The affine 0-Hecke monoid of type $A_1^{(1)}$ is $\mathcal{J}$ -trivial, so every two-sided cell contains exactly one left cell. Every staircase $(k, k-1, \dots, 1)$ is a 2-core with k distinct parts. Hence, for *every* bijection between cells and 2-cores, the formula fails at infinitely many non-empty cores. Everything below is formalised in Lean 4 with Mathlib, in a single file `Basic.lean`.

2 Reading and definitions

Definition 1 (Affine 0-Hecke monoid). Let $M = (m_{ij})$ be the Coxeter matrix of type $A_n^{(1)}$ on the nodes $\{0, \dots, n\}$ (cyclic Dynkin diagram): $m_{ii} = 1$; for $n \geq 2$, $m_{ij} = 3$ if $j \equiv i \pm 1 \pmod{n+1}$ and $m_{ij} = 2$ otherwise; for $n = 1$, $m_{01} = \infty$. The 0-Hecke monoid $H(M)$ is the monoid presented by generators $\pi_0, \dots, \pi_n$ and relations $\pi_i \pi_i = \pi_i$ (absorption) and, for $i \neq j$ with $m_{ij} < \infty$, the braid relation $\pi_i \pi_j \pi_i \cdots = \pi_j \pi_i \pi_j \cdots$ with m_{ij} factors on each side. In Lean: `affineA n : CoxeterMatrix (Fin (n+1))` (with ∞ encoded as 0, Mathlib’s convention), the relation `ZeroHeckeRel`, built from Mathlib’s `CoxeterSystem.alternatingWord`, and `AffineZeroHecke n := PresentedMonoid (ZeroHeckeRel (affineA n))`.

Definition 2 (Cells). In a monoid S , $x \mathcal{J} y$ iff $SxS = SyS$, i.e. $x = ayb$ and $y = cxd$ for some a, b, c, d ; and $x \mathcal{L} y$ iff $Sx = Sy$. These are Green’s relations; the principal two-sided ideal generated by a is $S^1 a S^1$ [Wikipedia, *Green’s relations*]. Following the conjecture’s own definition, a *two-sided cell* is a $\mathcal{J}$ -class and a *left cell* is an $\mathcal{L}$ -class. Since $\mathcal{L} \subseteq \mathcal{J}$, each left cell lies in a unique two-sided cell. The *number of left cells* of a two-sided cell C is the cardinality, in $\mathbb{N} \cup \{\infty\}$, of the set of $\mathcal{L}$ -classes contained in C . In Lean: `JRel`, `LRel` (both proved to be equivalence relations), `jClass`, `lClass`, `TwoSidedCell`, `numLeftCells` (a `Set.encard`).

Definition 3 (Hooks and cores). Partitions are Mathlib’s `YoungDiagrams`. The hook length of the cell (i, j) is `arm + leg + 1`: “the hook length is $a_\lambda(s) + l_\lambda(s) + 1$ ” [Wikipedia, *Young diagram*]

(hookLength). A partition is a t -core if none of its hook lengths is divisible by t (IsCore). For $t \geq 1$ this is equivalent to having no hook of length exactly t . One direction is trivial. For the other, use a β -set B of the partition: hooks of length ℓ correspond to pairs $b \in B, g \notin B$ with $b - g = \ell$. If $t \mid \ell$, walk along $b, b - t, b - 2t, \dots, g$; the first entry lies in B and the last does not, so some consecutive pair $b' \in B, b' - t \notin B$ occurs, giving a hook of length exactly t . Moreover, our refutation uses only the empty partition and the staircases $(k, \dots, 1)$, all of whose hook lengths are odd, so both definitions give the same 2-cores in every case used. The number of distinct parts is the number of distinct positive row lengths (distinctParts).

The hook-removal operation. “Removing all $(n + 1)$ -hooks” from a partition is modelled by an arbitrary map `strip`. Theorem 5 assumes only `strip( $\emptyset$ ) =  $\emptyset$` , which holds because the empty partition has no hooks at all. Theorem 9 assumes only that `strip` fixes 2-cores, which holds because a core has no hook of the given length to remove. Every reading of “removing all $(n + 1)$ -hooks” satisfies these assumptions. We do not need to fix a model of rim-hook removal.

The claim. `CellCoreDictionary n strip` states that there is a bijection e from the two-sided cells of `AffineZeroHecke n` to the $(n + 1)$ -cores with $\#\{\text{left cells in } C\} = \#\{\text{distinct parts of } \text{strip}(e(C))\}$ for every cell C .

3 The refutation

Lemma 4. *In any monoid, every two-sided cell contains at least one left cell (`numLeftCells_ne_zero`).*

Proof. If C is the $\mathcal{J}$ -class of x , then the $\mathcal{L}$ -class of x is contained in C , because $x = ay$ and $y = cx$ imply $x = ay1$ and $y = cx1$. $\square$

Theorem 5 (`not_cellCoreDictionary`). *For every n and every `strip` with `strip( $\emptyset$ ) =  $\emptyset$` , the conjecture `CellCoreDictionary n strip` is false.*

Proof. The empty partition $\emptyset$ has no cells, so it is an $(n + 1)$ -core. If e were such a bijection, let $C = e^{-1}(\emptyset)$. Then the number of left cells in C would equal the number of distinct parts of `strip( $\emptyset$ ) =  $\emptyset$` , which is 0. This contradicts Lemma 4. $\square$

This argument uses nothing about the monoid beyond Lemma 4. A reviewer may regard the empty core as degenerate, so we give a second refutation that avoids it.

Lemma 6 (`affineA1_jTrivial`). *The affine 0-Hecke monoid of type $A_1^{(1)}$ is $\mathcal{J}$ -trivial.*

Proof. Here $H = \langle \pi_0, \pi_1 \mid \pi_0^2 = \pi_0, \pi_1^2 = \pi_1 \rangle$, since $m_{01} = \infty$ gives no braid relation. Let π_i act on words over $\{0, 1\}$ by prepending i unless the word already starts with i (`act`). The relations hold for these maps, giving a monoid homomorphism $\Phi : H \rightarrow \text{End}(\{0, 1\}^*)$ (`actHom`). Put $\text{ev}(m) = \Phi(m)(\emptyset)$, the alternating normal form. The map sending a word to the product of its generators is a left inverse of ev , by induction using $\pi_i \pi_i = \pi_i$. Hence ev is injective (`eval_injective`). Let $\ell(m)$ be the length of $\text{ev}(m)$; note that $\text{ev}(pq) = \Phi(p)(\text{ev}(q))$. Each π_i either fixes a word or lengthens it. So $\ell(pq) \geq \ell(q)$, with equality only if $pq = q$ (`actHom_length`). By induction, $\text{ev}(p)$ is a prefix of $\Phi(p)(w)$ for every w (`eval_prefix`). So $\ell(pq) \geq \ell(p)$, with equality only if $pq = p$. Now suppose $x = ayb$ and $y = cxd$. Then $\ell(y) \leq \ell(ay) \leq \ell(ayb) = \ell(x) \leq \ell(cx) \leq \ell(cxd) = \ell(y)$. So $ay = y$, then $yb = y$, and therefore $x = y$. $\square$

Corollary 7 (`numLeftCells_affineA1`). For $A_1^{(1)}$, every two-sided cell is a singleton and contains exactly one left cell.

Lemma 8 (`staircase_isCore`, `distinctParts_staircase`). The staircase $\delta_k = (k, k-1, \dots, 1)$ (*staircase* k , cells $\{(i, j) : i + j < k\}$) is a 2-core with exactly k distinct parts.

Proof. Row i has length $k - i$ and column j has length $k - j$. So the hook length at (i, j) is $2(k - i - j) - 1$, which is odd. The row lengths $k, k-1, \dots, 1$ are distinct. $\square$

Theorem 9 (`affineA1_failures_infinite`, `not_cellCoreDictionary_one`). Let *strip* fix every 2-core. For every bijection e from the two-sided cells of the affine 0-Hecke monoid of type $A_1^{(1)}$ to the 2-cores, the set of 2-cores κ with $\#\{\text{left cells in } e^{-1}(\kappa)\} \neq \#\{\text{distinct parts of } \text{strip}(\kappa)\}$ is infinite. It contains every δ_k with $k \geq 2$. In particular *CellCoreDictionary 1 strip* is false.

Proof. The left-hand side is always 1, and the right-hand side at δ_{k+2} is $k+2$. The map $k \mapsto \delta_{k+2}$ is injective, because δ_{k+2} has $k+2$ distinct parts. $\square$

Remark 10. Lemma 4 holds for any notion of cells in which left cells partition two-sided cells. So Theorem 5 does not depend on reading “cells” as Green’s classes.

4 Lean formalisation and axiom audit

Lean 4 (leanprover/lean4:v4.33.1) with Mathlib v4.33.1; file `lean/Conjecture3848/Basic.lean` (369 lines, only import `Mathlib`). Main statements:

```
theorem not_cellCoreDictionary (n : Nat)
  (strip : YoungDiagram -> YoungDiagram) (hstrip : strip bot = bot) :
  not (CellCoreDictionary n strip)
theorem affineA1_jTrivial (x y : AffineZeroHecke 1) (h : JRel x y) :
  x = y
theorem affineA1_failures_infinite (strip : YoungDiagram -> YoungDiagram)
  (hstrip : forall mu, IsCore 2 mu -> strip mu = mu)
  (e : TwoSidedCell (AffineZeroHecke 1) ~> Core 2) :
  {k : Core 2 | numLeftCells (e.symm k).1
    != (distinctParts (strip k.1) : ENat)}.Infinite
theorem not_cellCoreDictionary_one (strip : YoungDiagram -> YoungDiagram)
  (hstrip : forall mu, IsCore 2 mu -> strip mu = mu) :
  not (CellCoreDictionary 1 strip)
```

(Shown in ASCII; the file uses the Unicode symbols.) `lake build` succeeds. `#print axioms` for each of the four theorems reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

Credit. The Green’s-relation set-up follows the accepted solution of the finite analogue 00000003843 (orionsheep, GPL-3.0), generalised here to arbitrary monoids.

References

- [1] Wikipedia, *Green’s relations*, https://en.wikipedia.org/wiki/Green%27s_relations.
- [2] Wikipedia, *Young diagram*, https://en.wikipedia.org/wiki/Young_diagram.